

Assessment of Implicit Large-Eddy Simulation with a Conservative Immersed Interface Method for turbulent cylinder flow

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ABSTRACT

The success of Large-Eddy Simulations (LES) of wall-bounded turbulence depends strongly on an accurate representation of the flow near the boundaries. Since in implicit LES the truncation error of the numerical discretization itself functions as SGS model, the order of accuracy of the discretization should be maintained near the boundary. In this paper, we analyze the performance of implicit LES for predicting turbulent flows along complex geometries. For the representation of wall-boundaries two different approaches are compared: (A) the case of body-fitted grids, and (B) a novel second-order accurate Conservative Immersed Interface Method (CIIM) for representing boundaries of arbitrary shape on Cartesian grids. The need for mapping domain boundaries on the grid raises the question whether a comparable accuracy with body-fitted grids can be achieved. In both (A) and (B) we employ the Adaptive Local Deconvolution Method (ALDM) for implicit LES. The assessment is carried out on the example of the flow over a circular cylinder at $Re = 3900$. Two simulations with comparable numerical resolution are performed and compared to experimental reference data. Both methods result in similar accuracy for turbulent statistics and mean integral quantities. This demonstrates that ALDM on Cartesian grids is a feasible and efficient approach to simulate turbulent flows on complex domains.

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1. Introduction

Implicit Large-Eddy Simulation (ILES) has shown considerable potential for the efficient representation of physically complex flows, see e.g. Hickel et al. (2007a) and Hickel and Adams (2008) and references therein. In ILES the truncation error of the discretization of the convective terms acts as a subgrid-scale model which is thus an inherent feature of the discretization (Adams et al., 2004). In this context, the Adaptive Local Deconvolution Method (ALDM) (Hickel et al., 2006; Hickel and Adams, 2007; Hickel et al., 2007b) uses a discretization based on solution-adaptive deconvolution where parameters are calibrated in such a way that the truncation error properly models the subgrid-scale energy transfer. Applications to canonical configurations, such as isotropic turbulence (Hickel et al., 2006), plane channel flow (Hickel and Adams, 2007), and turbulent boundary layer separation (Hickel and Adams, 2008) show excellent agreement with the reference data from theory, experiment, or direct numerical simulations. Given this performance, our motivation is to use ALDM for LES of flows around complex geometries.

ALDM has been developed primarily for Cartesian grids due to the superior computational efficiency and because Cartesian grids

allow for an easy control of the truncation error. Bounding surfaces of the flow that are not aligned with the grid have to be accounted for by an Immersed Interface (II) approach, for which discrete mass and momentum conservation and high-order of accuracy near the interfaces pose a challenge. For obtaining physically realistic numerical solutions discrete mass conservation (Arakawa, 1966; Cheny and Botella, 2009; Morinishi et al., 1998; Verstappen and Veldman, 2003) and sharp representation of the fluid–solid interface are essential for LES of high-Reynolds number flows. An overview on existing Immersed Boundary (IB) approaches is given in Mittal and Iaccarino (2005). Note that II methods are a generalization of immersed boundary methods as the formulation can be applied also to fluidic interfaces.

Recently, ALDM has been successfully translated to general curvilinear coordinates and implemented in a finite-volume solver for the incompressible Navier–Stokes equations on three-dimensional block-structured, body-fitted grids (Hickel et al., 2009). Boundary conditions are imposed in a natural way on body-fitted grids, which facilitate accurate wall-layer numerical predictions. However, the generation of good-quality curvilinear grids for highly complex geometries is time-consuming and difficult. Contradictory requirements, namely adequate local resolution and minimum number of grid points can deteriorate the grid quality and therefore adversely affect accuracy and numerical convergence properties. This is the main reason why II methods are more desirable for

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complex domains despite the superior boundary representation of body-fitted methods.

In this paper we assess the accuracy and the performance of a novel second-order, Conservative Immersed Interface Method (CIIM) (Meyer et al., submitted for publication) in the framework of ILES. The objective of this investigation is to qualify whether CIIM can represent turbulent flows in the vicinity of immersed interfaces with sufficient accuracy. As test case, we consider the flow over a circular cylinder at Reynolds number $Re = 3900$ based on the cylinder diameter and the free-stream velocity. This case is widespread in the literature and, despite the variation of published results and sometimes not very clearly specified configurations, can be considered as established reference. At this Reynolds number the separated laminar shear layers undergo transition in the near wake of the cylinder. An accurate representation of the flow at the immersed interface is crucial for a correct prediction of the global flow physics, making this test case suitable for the assessment of CIIM. Many Direct Numerical Simulations (Dröge, 2007; Ma et al., 2000) and Large-Eddy Simulations (Breuer, 1998; Fröhlich et al., 1998; Kravchenko and Moin, 2000; Franke and Frank, 2002; Mahesh et al., 2004; Park et al., 2004; Parnaudeau et al., 2008) have been conducted for this flow. In order to address the particular concern of accurate boundary treatment, we compare the results of two implicit LES with ALDM, one on a curvilinear body-fitted grid with a usual formulation of the wall boundary condition, and one on a Cartesian grid with CIIM. Both computations are also compared with the pressure measurements of Norberg published in Kravchenko and Moin (2000), the particle-image velocimetry data of Lourenco and Shih (extracted from Dröge (2007)), the hot-wire measurements of Ong and Wallace (1996) and the particle-image velocimetry data of Parnaudeau et al. (2008).

In Section 2 we summarize the underlying spatial and temporal discretization of the governing equations used by the two flow solvers employed in this paper. Computational results are presented and analyzed in Section 3. Concluding remarks are given in Section 4.

2. Numerical method

2.1. Underlying finite-volume discretization and ALDM

We consider a generic conservation equation in differential form on the domain Ω

$$\partial_t \mathbf{Q} + \nabla \cdot \mathbf{F} = 0, \quad (1)$$

with appropriate boundary conditions on $\partial\Omega$. The vector $\mathbf{Q}$ represents the conserved quantities and $\mathbf{F}$ is a flux function. The Navier–Stokes equations for incompressible fluid flow with constant density are recovered when $\mathbf{Q}$ equals the velocity $\mathbf{u} = [u, v, w]$ and $\mathbf{F}$ is the nonlinear flux function

$$\mathbf{F} = \mathbf{u}\mathbf{u} + \mathbf{I}p - \nu\nabla\mathbf{u}, \quad (2)$$

where ν represents the kinematic viscosity, p is the pressure and $\mathbf{I}$ is the unity tensor. Note that the non-dimensional constant density is set to $\rho = 1$. Mass conservation on Ω is ensured by the incompressible continuity equation

$$\nabla \cdot \mathbf{u} = 0. \quad (3)$$

Integrating Eq. (1) over the volume $V_{i,j,k} \cap \Omega$ of a computational cell (i,j,k) and a time step t_n to t_{n+1} with time step size $\Delta t = t_{n+1} - t_n$ yields

$$\frac{1}{V_{i,j,k}\Delta t} \int_{t_n}^{t_{n+1}} dt \int_{V_{i,j,k} \cap \Omega} dx dy dz \left(\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{F} \right) = 0. \quad (4)$$

A finite-volume discretization of Eq. (4) is obtained by

$$\frac{1}{V_{i,j,k}\Delta t} \int_{t_n}^{t_{n+1}} dt \left(\int_{V_{i,j,k} \cap \Omega} dx dy dz \frac{\partial \mathbf{u}}{\partial t} + \int_{\partial(V_{i,j,k} \cap \Omega)} dS \mathbf{F} \cdot \mathbf{n}_\perp \right) = 0, \quad (5)$$

where $\mathbf{n}_\perp$ is the normal vector and dS are the cell faces. In this paper, a fractional-step method is used and fluxes in Eq. (5) are computed with the Adaptive Local Deconvolution Method (ALDM).

The theoretical background of implicit LES with ALDM (Adams et al., 2004; Hickel et al., 2006) follows from the formal equivalence between cell-averaging and reconstruction in finite-volume discretizations, and top-hat filtering and deconvolution in explicit SGS-modeling. With ALDM, a local reconstruction of the unfiltered solution is obtained from a solution-adaptive combination of approximation polynomials. Deconvolution and interpolation are done simultaneously using Lagrangian polynomials as proposed by Harten (1987). Deconvolution is regularized by limiting the degree of local approximation polynomials and by permitting all polynomials up to a certain degree to contribute to the approximately deconvolved solution. Adaptivity of the deconvolution operator is achieved by dynamically weighing the respective contributions. A suitable consistent numerical flux function operates on the approximately deconvolved solution. The solution-adaptive stencil-selection scheme and the numerical flux function contain free parameters which can be used to adjust the spatial truncation error of the discretization.

Free parameters of ALDM are selected such that the truncation error of the discretization functions as physically motivated SGS model. Given that the primary purpose of an SGS model is to provide the correct spectral distribution of the dissipation of resolved scales by interactions with modeled SGS stresses, free discretization parameters were calibrated by constraining the numerical dissipation to the physical SGS dissipation obtained from the analysis of nonlinear interactions in homogeneous isotropic turbulence (Hickel et al., 2006). The general validity of the determined model is demonstrated for various canonical flow configurations (Hickel et al., 2006; Hickel and Adams, 2007; Hickel and Adams, 2008).

In LES the chosen grid size essentially defines the smallest represented physical scale. At finite grid size, the numerical truncation errors interfere with the subgrid-scale model and modeling errors. In particular for implicit LES, numerical discretization and turbulence model are indistinguishable. For non-turbulent flows ALDM corresponds to a standard central second-order finite-volume scheme, whereas for turbulent flows it provides a subgrid-scale model implicitly.

2.2. Curvilinear grid simulation code LESOCC2

The simulation code Large-Eddy Simulation on Curvilinear Coordinates (LESOCC2) (Hinterberger, 2004) employs non-orthogonal, boundary-fitted, block-structured grids. Boundary-fitted block-structured grids can offer considerable computational advantages for predicting wall-bounded turbulence: Imposing no-slip wall boundary conditions is straightforward for grid-aligned domain boundaries. The algorithm uses a cell-centered variable arrangement and Cartesian velocity components. The convective fluxes are discretized by ALDM in an algorithmically simplified version (SALD) (Hickel and Adams, 2006). SALD preserves the modeling capability while computational costs are reduced significantly. Another advantage of SALD is that it allows for a straightforward application to curvilinear grids as the reconstruction of the unfiltered solution at cell faces by Harten-type deconvolution polynomials can be done dimension by dimension (Hickel et al., 2009). Viscous fluxes are discretized by centered differences.

Numerical stability for the collocated grid arrangement is maintained by employing a momentum–interpolation technique.

2.3. Cartesian grid simulation code INCA

The same implicit LES method has been implemented for the Incompressible Navier–Stokes equations on Cartesian Adaptive staggered grids (INCA). Wall boundary conditions are imposed on the Cartesian grid by a second-order Conservative Immersed Interface Method (CIIM) (Meyer et al., submitted for publication). CIIM is based on the finite-volume discretization Eq. (5) applied to all cells of the entire computational domain, and a modification for cells that are cut by the immersed interface. The main building blocks of CIIM are outlined in the following subsections. For a more details refer to Meyer et al. (submitted for publication).

2.3.1. Cut cell volume balance

The intersection between the immersed boundary and the finite-volume cell is approximated as planar interface separating fluid and solid part (see Fig. 1). The volume fraction of the fluid part of a cut cell is denoted $\alpha_{i,j,k}$, with $0 \leq \alpha_{i,j,k} \leq 1$. The volume $V_{i,j,k}$ of a cut cell (i,j,k) can be expressed as $V_{i,j,k} = \alpha_{i,j,k} \Delta x \Delta y \Delta z$, where Δx , Δy and Δz are the cell sizes in the respective coordinate directions. The cell faces S are represented by two parts: one is the combination of the six segments of cell faces cut by the interface, which can be written in the form of $A_{i\pm 1/2,j,k} \Delta y \Delta z$, $A_{i,j\pm 1/2,k} \Delta x \Delta z$, and $A_{i,j,k\pm 1/2} \Delta x \Delta y$, where $1 \leq A_{k,l,m} \leq 0$ is the face aperture, the other is the part of the interface $\Gamma_{i,j,k}$ inside of the cell (i,j,k) . For a cut cell the general formulation of Eq. (5) yields

$$\begin{aligned} \mathbf{u}_{i,j,k}^{n+1} &= \mathbf{u}_{i,j,k}^n \\ &+ \frac{\Delta t [A_{i+1/2,j,k} \mathbf{F}_{i+1/2,j,k} - A_{i-1/2,j,k} \mathbf{F}_{i-1/2,j,k}]}{\alpha_{i,j,k} \Delta x} \\ &+ \frac{\Delta t [A_{i,j+1/2,k} \mathbf{F}_{i,j+1/2,k} - A_{i,j-1/2,k} \mathbf{F}_{i,j-1/2,k}]}{\alpha_{i,j,k} \Delta y} \\ &+ \frac{\Delta t [A_{i,j,k+1/2} \mathbf{F}_{i,j,k+1/2} - A_{i,j,k-1/2} \mathbf{F}_{i,j,k-1/2}]}{\alpha_{i,j,k} \Delta z} \\ &+ \frac{\Delta t (\mathbf{C} + \mathbf{D})}{\alpha_{i,j,k} \Delta x \Delta y \Delta z}, \end{aligned} \quad (6)$$

where the discretization in time by an Euler forward scheme shown here for brevity corresponds to a substep of the third-order Runge–Kutta scheme used in the simulations. $\mathbf{C}$ and $\mathbf{D}$ are additional terms to account for the influence of the immersed interface on the fluid. A sketch of a cut cell in two dimensions for clarity is given in Fig. 1, explanations will follow below. For full fluid cells it is $\mathbf{C} = \mathbf{D} = 0$, and Eq. (6) degenerates to a standard finite-volume discretization

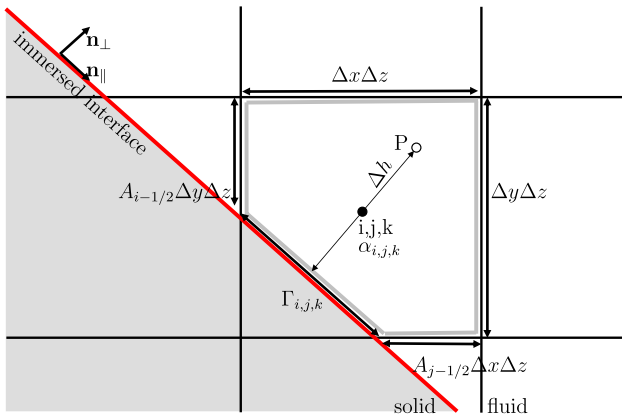


Fig. 1. Sketch of a cut cell (i,j,k) for the conservative immersed interface method.

on a Cartesian grid. Please note that the approach operates on fluxes only, so that momentum is discretely conserved.

2.3.2. Friction term $\mathbf{D}$

In order to account for the shear stress on the interface Γ , we add the friction force $\mathbf{D}$ to the flux balance of the cut cell (see Fig. 1). Using a linear approximation of the wall-normal velocity gradient we obtain

$$\mathbf{D} = - \int_{\Gamma} \tau \cdot \mathbf{n}_{\perp} dS = - \int_{\Gamma_{i,j,k}} \nu \frac{\partial \mathbf{u}_{\parallel}}{\partial \mathbf{n}_{\perp}} \cdot \mathbf{n}_{\perp} dS = - \frac{\Gamma \nu}{\Delta h} (\mathbf{u}_{\parallel} - \mathbf{u}_r), \quad (7)$$

where $\partial \mathbf{u}_{\parallel} / \partial \mathbf{n}_{\perp}$ only considers the tangential velocity $\mathbf{u}_{\parallel}$, which is determined by trilinear Lagrangian interpolation at point P , see Fig. 1, from the velocities at the eight neighboring velocity-cell centers. $\mathbf{u}_r$ is the known velocity at the boundary. The distance Δh needs to be related to the grid spacing so that a linear velocity gradient can be assumed for calculations resolving the viscous sublayer. We found that

$$\Delta h = \frac{\sqrt{(\Delta x \cdot n_{\perp,x})^2 + (\Delta y \cdot n_{\perp,y})^2 + (\Delta z \cdot n_{\perp,z})^2}}{2} \quad (8)$$

gives good results for all possible cut cell topologies. Δx , Δy and Δz are the dimensions of the control volume $V_{i,j,k}$ and $n_{\perp,x}$, $n_{\perp,y}$, $n_{\perp,z}$ are components of the normal vector on the interface $\Gamma_{i,j,k}$.

2.3.3. Coupling term $\mathbf{C}$

For impermeable non-moving boundaries the interface normal velocity $\mathbf{u}_{\perp} = (\mathbf{u} \cdot \mathbf{n}_{\perp}) \mathbf{n}_{\perp}$ of the fluid vanishes. Zero normal velocity is generally imposed in the discrete Poisson equation, see also Section 2.3.4. On a staggered grid some very small cut cells do not contribute to the discrete Poisson equation. For these cells we add a pressure–velocity coupling term

$$\mathbf{C} = -\mathbf{u}_{\perp,i,j,k} \frac{V_{i,j,k}}{\Delta t} \quad (9)$$

to the flux balance of Eq. (6).

2.3.4. Homogeneous Neumann condition for pressure

The solution algorithm is based on a fractional-step method. The pressure of the previous time step is used to predict an intermediate velocity field $\mathbf{u}^*$, which is not divergence free. The incompressibility constraint is satisfied by a pressure correction ϕ which is obtained from a Poisson equation with a homogeneous Neumann condition on the interface Γ . On a staggered grid, velocities are known at the cell faces so that the discretized Poisson equation can be written as

$$\begin{aligned} &\frac{1}{\Delta x} \left[A_{i+1/2,j,k} \frac{\partial \phi}{\partial x} \Big|_{i+1/2,j,k} - A_{i-1/2,j,k} \frac{\partial \phi}{\partial x} \Big|_{i-1/2,j,k} \right] \\ &+ \frac{1}{\Delta y} \left[A_{i,j+1/2,k} \frac{\partial \phi}{\partial y} \Big|_{i,j+1/2,k} - A_{i,j-1/2,k} \frac{\partial \phi}{\partial y} \Big|_{i,j-1/2,k} \right] \\ &+ \frac{1}{\Delta z} \left[A_{i,j,k+1/2} \frac{\partial \phi}{\partial z} \Big|_{i,j,k+1/2} - A_{i,j,k-1/2} \frac{\partial \phi}{\partial z} \Big|_{i,j,k-1/2} \right] - \frac{\alpha_{i,j,k} \Gamma_{i,j,k} r_{p,\Gamma}}{V_{i,j,k}} \\ &= \frac{1}{\Delta x} [A_{i+1/2,j,k} u_{i+1/2,j,k}^* - A_{i-1/2,j,k} u_{i-1/2,j,k}^*] \\ &+ \frac{1}{\Delta y} [A_{i,j+1/2,k} v_{i,j+1/2,k}^* - A_{i,j-1/2,k} v_{i,j-1/2,k}^*] \\ &+ \frac{1}{\Delta z} [A_{i,j,k+1/2} w_{i,j,k+1/2}^* - A_{i,j,k-1/2} w_{i,j,k-1/2}^*] - \frac{\alpha_{i,j,k} \Gamma_{i,j,k} r_{u,\Gamma}}{V_{i,j,k}}, \end{aligned} \quad (10)$$

where u^* , v^* , w^* are the intermediate velocities at the cell faces and $r_{u,\Gamma}$ and $r_{p,\Gamma}$ are the velocity and pressure interface condition at $\Gamma_{i,j,k}$. The pressure gradient is discretized with second-order central differences. For non-moving boundaries we set $r_{u,\Gamma} = \mathbf{u}_r \cdot \mathbf{n}_{\perp}$ to

zero. A homogeneous Neumann condition $\partial\phi/\partial\mathbf{n}_\perp = 0$ at the interface corresponds to $r_{p,r} = 0$. The pressure Poisson equation is solved only on the fluid domain with Neumann condition at the immersed interface. Mass is therefore conserved up to the convergence threshold, which can be set to machine precision if desired.

2.3.5. Mixing procedure

Cartesian cut cell methods, such as the one employed in this paper, inevitably generate very small fluid cells. A stable time integration based on the time step according to the full cell size CFL condition may not be possible without a special treatment of these cells.

In the present method, the fluid of the small cells is mixed with that of the neighboring cells. We consider cells with a volume fraction $\alpha_{i,j,k} \leq 0.5$ as small cells. Target cells are determined from an evaluation of the normal vector. Unlike Hu et al. (2006) we also consider diagonal cells, so that we have three target cells in two dimensions, and seven target cells in three dimensions. For each target cell we determine an exchange flux based on the cell sizes of the target cell and small cell. Once the exchange flux of all target cells is determined the intermediate solution is updated. Note that discrete conservation is ensured by a flux formulation, where the conservative quantity obtained by a small cell corresponds to a loss of a target cell. This mixing is carried out before the final pressure projection, so that the solution after each time step is divergence free. This mixing procedure has no significant effect on the overall accuracy of CIIM. Meyer et al. (submitted for publication) have proven second-order grid convergence so that no interference with the ALDM turbulence model is expected.

3. Assessment of CIIM

3.1. Test case

The flow over a circular cylinder at $Re = 3900$ (based on cylinder diameter d , the reference density ρ_∞ and freestream velocity U_∞) is a challenging test case for LES in general and for CIIM in particular. At this Reynolds number laminar shear layers separate from the cylinder and undergo transition in the near wake of the cylinder. A good representation of flow physics in the vicinity of the cylinder surface is necessary to correctly predict the separation of the shear layers. Both, the transition region and the turbulent wake region require sufficient grid resolution for capturing turbulence phenomena.

3.2. Computational setup

The computational domain is shown in Fig. 2. Free-slip boundary conditions are applied at the upper and lower side and periodicity is assumed in spanwise direction. At the inlet a uniform velocity profile and at the outlet a pressure condition is defined.

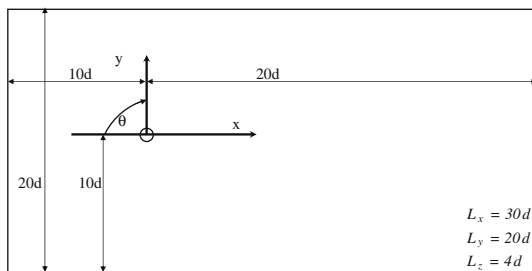


Fig. 2. Computational setup for the flow over a circular cylinder at $Re = 3900$ (θ denotes the circumferential coordinate; L_x , L_y , L_z are the domain extents in x -, y - and z -direction).

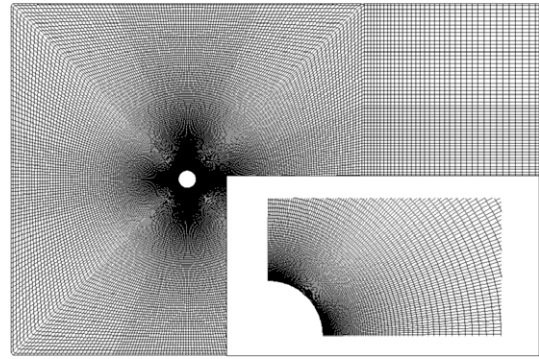


Fig. 3. Curvilinear grid with O-block topology is shown for the flow over a circular cylinder at $Re = 3900$; right corner: zoom into the cylinder region.

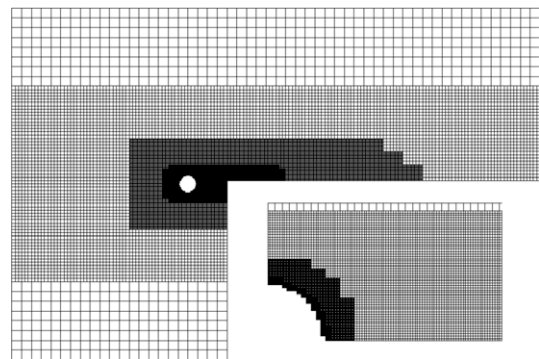


Fig. 4. Every third computational cell of a locally-refined Cartesian grid is shown for the flow over a circular cylinder at $Re = 3900$; right corner: zoom into the cylinder region.

At the boundary of the obstacle a viscous-wall boundary condition is prescribed for the calculation on the curvilinear grid, and for the calculation on the Cartesian grid the immersed interface condition is imposed by CIIM. An O-H topology is used for the curvilinear grid with a total number of 6 million computational cells, see Fig. 3. The Cartesian grid has a total number of 6.5 million computational cells and is locally refined at the cylinder, see Fig. 4. Both grids resolve the viscous sublayer at the cylinder with a grid spacing of $0.0025d$. Both grids further have a homogeneous spanwise resolution of $0.05d$. The LES of Franke and Frank (2002) and Fröhlich et al. (1998) use similar spatial resolution for this flow.

For the discretization in space, ALDM is used for the convective terms, and a second-order accurate centered scheme for the viscous terms and the pressure gradient. Time advancement is performed using a third-order explicit three-step Runge-Kutta scheme (Shu, 1988). The pressure Poisson equation is solved at every Runge-Kutta substep. Computations are carried out with a CFL number of 1.0 based on the full cells on the Cartesian grid and with $CFL = 0.7$ on the curvilinear grid.

3.3. Reference data

The pressure measurements of Norberg (data extracted from Kravchenko and Moin (2000)) provide the only experimental surface pressure data known to the authors. Turbulent statistics in the very near wake of the cylinder have been measured by Lourenco and Shih (1993) using PIV. These data are commonly used in numerical studies for validation (e.g. Franke and Frank, 2002; Dröge, 2007) and investigation (e.g. Kravchenko and Moin, 2000; Ma et al., 2000; Parnaudeau et al., 2008), and yet its validity in

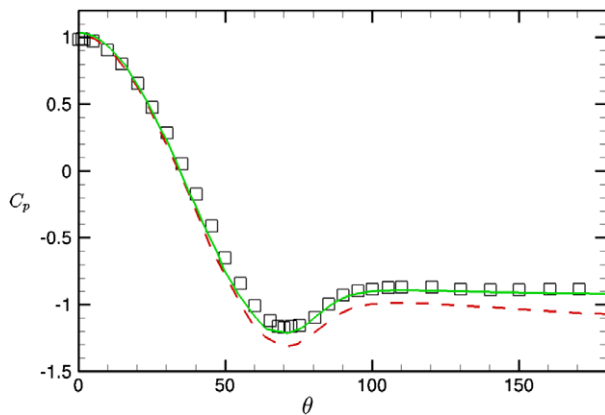


Fig. 5. The pressure coefficient C_p along the surface of a circular cylinder at $Re = 3900$ ($\theta = 0^\circ$ in the stagnation point, $\square$ experimental data of Norberg at $Re = 4020$, published in Kravchenko and Moin (2000), --: LESOCC2 and -: INCA).

the recirculation zone often is subject to discussion. Beaudan and Moin (1994) attribute discrepancies in the velocity profiles to asymmetric experimental conditions. This disagreement is confirmed by Mittal and Moin (1997) as well as Kravchenko and Moin (2000). Kravchenko and Moin (2000) further argue that observed differences can be attributed to transitional effects in the separating shear layers of the experiment. Norberg (1994) showed that aspect ratios of 60–70 are required for obtaining results unaffected by cylinder length under experimental conditions. Szepessy and Bearman (1992) report differences in the vortex formation length for different aspect ratios. In view of these findings the aspect ratio of 21 in the experiments of Lourenco and Shih (1993) appears as rather small.

However, Parnaudeau et al. (2008) whose PIV data we also use for comparison have the same aspect ratio in their experiment as Lourenco and Shih (1993) but obtain different results. Furthermore, the PIV data of the experiments of Parnaudeau et al. (2008) for mean velocity on the wake centerline are consistent with the hot-wire measurements of Ong and Wallace (1996) but are inconsistent with their own hot-wire data. Due to these discrepancies, we consider both the PIV data of Lourenco and Shih (1993) and Parnaudeau et al. (2008) for the very near wake and the hot-wire measurements of Ong and Wallace (1996) for the near wake comparisons.

3.4. Results

For a quantitative validation, turbulence statistics are sampled over 60 shedding cycles after steady vortex shedding was established. The results are compared with pressure measurements of Norberg (data extracted from Kravchenko and Moin (2000)), the Particle-Image Velocimetry (PIV) data of Lourenco and Shih (1993) (extracted from Dröge (2007)), the hot-wire measurements of Ong and Wallace (1996) and the PIV data of Parnaudeau et al. (2008).

3.4.1. Pressure coefficient C_p

The pressure coefficient $C_p = 0.5(p - p_\infty)/\rho_\infty U_\infty^2$ is a relevant contribution to the total drag. The reference pressure p_∞ is the time-averaged pressure at the inlet plane. C_p profiles for both LES, see Fig. 5, show good agreement with the experiment of Norberg, published in Kravchenko and Moin (2000). Note that the calculation with CIIM shows a smooth C_p profile unlike that often reported for discrete forcing approaches (Chen and Botella, 2009; Kang et al., 2009; Muldoon and Acharya, 2008).

Table 1

Drag coefficient, mean base coefficient, separation angle, normalized recirculation bubble length and Strouhal number for the flow over a circular cylinder at $Re = 3900$.

Case	C_D	$C_{p,b}$	θ_{sep}	L_r/d	St
Cardell (1993), experiment	–	–	–	1.33	0.215
Lourenco and Shih (1993), experiment	0.99	–	86°	1.19	0.215
Norberg (1994), experiment	0.98	–0.9	–	–	–
Ong and Wallace (1996), experiment	–	–	–	–	0.21
Dong et al. (2006), experiment	–	–	–	1.47	0.203
Parnaudeau et al. (2008), experiment	–	–	–	1.51	0.208
Ma (2000), DNS	0.96	–0.96	89.1°	1.12	0.203
Dong et al. (2006), DNS	–	–	–	1.36	0.203
Dröge (2007), DNS	1.01	–0.88	87.7°	1.26	0.210
Fröhlich et al. (1998), LES	1.08	–1.03	88.1°	1.09	0.216
Franke and Frank (2002), LES	0.99	–0.85	88.2°	1.64	0.209
Kravchenko and Moin (2000), LES	1.04	–0.94	88°	1.35	0.210
Parnaudeau et al. (2008), LES	–	–	–	1.56	0.208
Present curvilinear grid, LES	1.07	–1.05	88.9°	1.18	0.215
Present Cartesian grid, LES	1.05	–0.92	88.0°	1.38	0.210

3.4.2. Mean integral quantities

We compare the mean drag coefficient, base suction coefficient, mean separation angle, mean recirculation length and Strouhal number with the literature in Table 1. The time-averaged drag coefficient per unit length is $C_D = 0.5F_D/\rho_\infty U_\infty^2 S$, where S is the projected cylinder surface in streamwise direction and F_D is the force in streamwise direction and consists of the friction force F_{Df} and the pressure force F_{Dp} acting in the streamwise direction at the boundary of the obstacle. The base suction coefficient $C_{p,b}$ is the pressure coefficient on the cylinder at $\theta = 180^\circ$. The mean separation angle θ_{sep} is determined from the condition of vanishing wall shear stress. The normalized length of recirculation bubble is denoted by L_r/d . The Strouhal number is defined to be $St = fd/U_\infty$, where f is the shedding frequency. f is determined from a Fourier transform of a streamwise velocity time signal obtained at $x/d = 2$, $y/d = 1$.

All presented data for the drag coefficient are in good agreement. Ma et al. (2000) reported the lowest drag coefficient with 0.96 and Fröhlich et al. (1998) obtained the highest with 1.07. In our simulation on the curvilinear grid we obtain a similar value as the LES of Fröhlich et al. (1998) who also used LESOCC2 but in combination with a dynamic Smagorinsky model. The drag coefficient obtained on the Cartesian grid is close to the value obtained in the LES of Kravchenko and Moin (2000).

The base suction coefficient is strongly related to the drag coefficient, since drag is mainly due to pressure drag and the pressure varies mostly at $\theta = 180^\circ$ for all simulations and experiments. The results on the curvilinear grid are again close to the results obtained by Fröhlich et al. (1998). The results obtained on the Cartesian grid are close to the experimental data of Norberg (1994), see also Fig. 5.

The separation angle in the experiment by Lourenco and Shih (1993) is considerably lower than the angles of the simulations. However the angles found in the simulations including ours are all in good agreement. The difference for the recirculation bubble length L_r/d for the experiments and simulations is significant. Fröhlich et al. (1998) and Ma et al. (2000) and our calculation on the curvilinear grid have the smallest recirculation bubble, which is also in rather good agreement with the experimental results of Lourenco and Shih (1993). The largest recirculation length is reported by Franke and Frank (2002) for their LES. The results obtained on the Cartesian grid are close to the experimental results of Cardell (1993), the results obtained in the DNS of Dong et al. (2006) and the results obtained in the LES of Kravchenko and Moin (2000).

The results for the Strouhal number in the simulations are all within the range of the experimental data, only Ma et al. (2000)

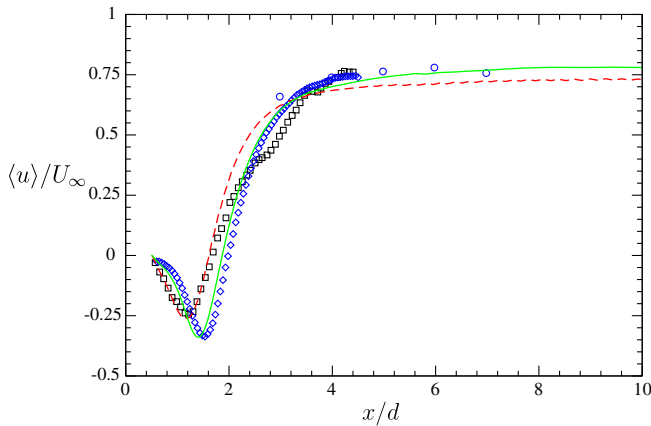


Fig. 6. Mean streamwise velocity on the centerline of the cylinder for the flow over a circular cylinder at $Re = 3900$ ($\square$: experimental data Lourenco and Shih (extracted from Dröge (2007)), $\diamond$: experimental data (Parnaudeau et al., 2008), $\circ$: experimental data (Ong and Wallace, 1996), --: LESOCC2 and -.: INCA).

and Dong et al. (2006) reported results that are lower. Our results match the experimental results of Cardell (1993) and Lourenco and Shih (1993).

3.4.3. First and second-order statistics

The mean streamwise velocity along the centerline is shown in Fig. 6. We clearly see that the recirculation zone in the experiments of Lourenco and Shih (1993) is shorter than in the experiments of Parnaudeau et al. (2008). The calculation on the curvilinear grid shows good agreement with the results of Lourenco and Shih (1993) while the calculation on the Cartesian grid shows good agreement with the experimental results of Parnaudeau et al. (2008). The dip at $x/d = 3$ of the experiments of Lourenco and Shih (1993) is not reproduced neither by the experiments of Parnaudeau et al. (2008) nor by any simulation reported in the literature.

The Figs. 7–10 compare the statistics for the very near wake of the cylinder with the experiments of Lourenco and Shih (1993) and Parnaudeau et al. (2008). Fig. 7 shows velocity profiles at four different downstream locations ($x/d = 0.58, 1.06, 1.54, 2.02$) for the mean streamwise velocity $\langle u \rangle / U_\infty$. A strong velocity deficit occurs in the region of the recirculation bubble. The mean velocity-profile shows a U-shape close to the cylinder which evolves towards a V-shape further downstream. At $x/d = 0.58$ both implicit LES and the data of Lourenco and Shih are in good agreement. Please note the different shape of the profiles at $x/d = 1.06$ for the two experiments. The results of Lourenco and Shih show a V-shape whereas Parnaudeau et al. (2008) show a U-shape profile. Both shapes have been obtained also in numerical studies. A V-shape profile at $x/d = 1.06$ along with a longer recirculation length has been observed by, e.g. Kravchenko and Moin (2000) and Parnaudeau et al. (2008) and also for the present implicit LES on the Cartesian grid. On the other hand, our calculation on the curvilinear grid agrees with the results of Lourenco and Shih (1993) and the LES of Fröhlich et al. (1998) and Fröhlich et al. (2006). Ma et al. (2000) were able to reproduce both shapes by DNS, where doubling the spanwise extent of the domain changes the mean profile from U-shape to V-shape. Results for the mean transversal velocity $\langle v \rangle / U_\infty$ are shown in Fig. 8. It should be mentioned that profiles of Lourenco and Shih are shifted such that $\langle v \rangle = 0$ for large y/d is recovered, as done also by Franke and Frank (2002). This anomalous behavior in the data of Lourenco and Shih can be attributed to experimental disturbances (Parnaudeau et al., 2008). Despite the shifting, the calculations in general agree better with the data of Parnaudeau et al. (2008). Possible explanations are discussed below.

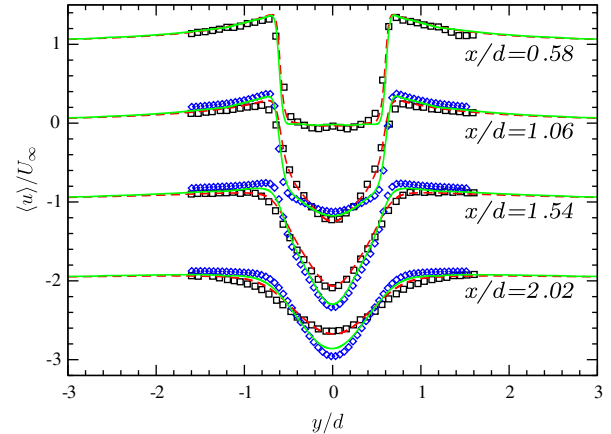


Fig. 7. Mean streamwise velocity at different locations in the wake of a circular cylinder at $Re = 3900$ ($\square$: experimental data Lourenco and Shih (extracted from Dröge (2007)), $\diamond$: experimental data (Parnaudeau et al., 2008), --: LESOCC2 and -.: INCA).

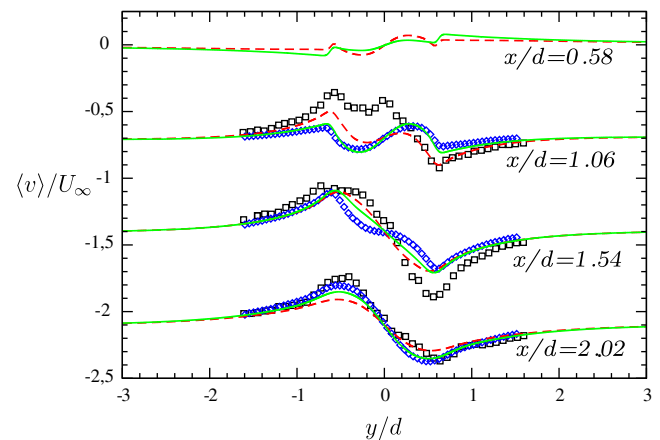


Fig. 8. Mean transverse velocity at different locations in the wake of a circular cylinder at $Re = 3900$ ($\square$: experimental data Lourenco and Shih (extracted from Dröge (2007)), $\diamond$: experimental data (Parnaudeau et al., 2008), --: LESOCC2 and -.: INCA).

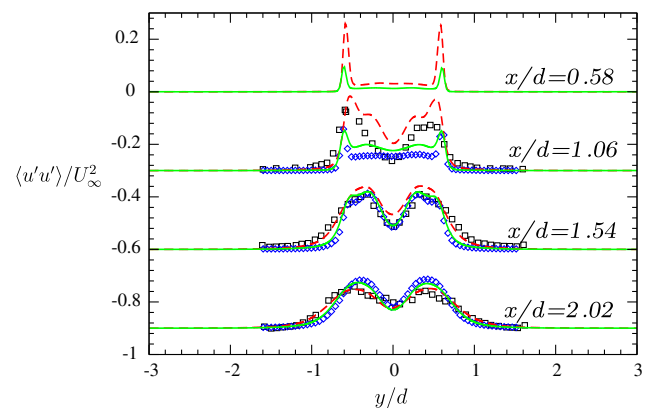


Fig. 9. Streamwise velocity fluctuations at different locations in the wake of a circular cylinder at $Re = 3900$ ($\square$: experimental data Lourenco and Shih (extracted from Dröge (2007)), $\diamond$: experimental data (Parnaudeau et al., 2008), --: LESOCC2 and -.: INCA).

In Fig. 9 the Reynolds normal stresses $\langle u'u' \rangle / U_\infty^2$ are shown. At $x/d = 0.58$ the calculation on the curvilinear grid shows higher peaks in the Reynolds normal stresses than the calculation on

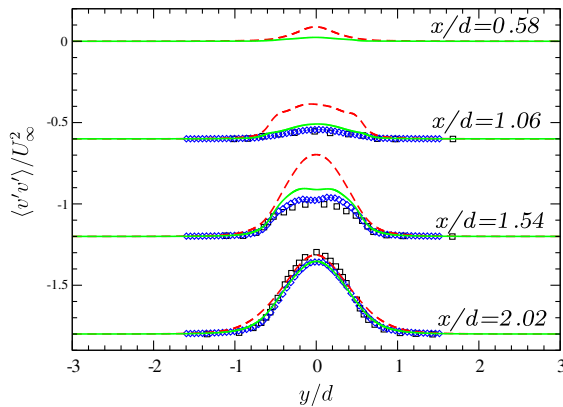


Fig. 10. Cross-flow velocity fluctuations at different locations in the wake of a circular cylinder at $Re = 3900$ ($\square$: experimental data Lourenco and Shih (extracted from Dröge (2007)), $\diamond$: experimental data (Parnaudeau et al., 2008), --: LESOCC2 and -: INCA).

the Cartesian grid. The two sharp peaks at $x/d = 1.06$ can be attributed to the transitional state of the shear layers, which show a flapping behavior due to primary vortex formation. The two peaks can be matched in magnitude and position by our implicit LES on the Cartesian grid. The results on the curvilinear grid again show an overshoot with the respect to other data. Better agreement is reached further downstream for all data. For the cross-flow normal Reynolds stresses $\langle v'v' \rangle / U_\infty^2$ the calculation on the Cartesian grid shows reasonable agreement with both experiments, see Fig. 10, while the calculation on the curvilinear grid shows an overshoot at $x/d = 0.58, 1.06, 1.54$. The overshoot in the Reynolds stresses of the calculation on the curvilinear grid implies that with respect to the Cartesian-grid calculation laminar-turbulent transition of the separated shear layers occurs closer to the cylinder, which leads to a shorter length of the recirculation bubble. This is also confirmed by the contour plots shown in Figs. 15 and 16). The mean velocity fields in Fig. 15 are similar apart from the size of the recirculation bubble, whereas the velocity fluctuations in Fig. 16) clearly show earlier transition in the detached shear layers of our LESOCC2 calculation.

Figs. 11–14 compare the statistics for the near wake ($x/d = 4, 6, 7, 10$) with the experiments of Ong and Wallace (1996). Fig. 11 shows the mean streamwise velocity at four different locations. The slightly lower velocity at $x/d = 4$ and $x/d = 6$ on the wake centerline indicates that the recirculation length in the

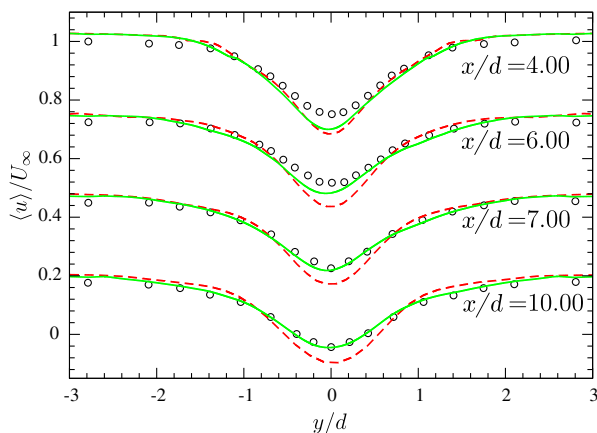


Fig. 11. Mean streamwise velocity at different locations in the wake of a circular cylinder at $Re = 3900$ ($\circ$: experimental data (Ong and Wallace, 1996), --: LESOCC2 and -: INCA).

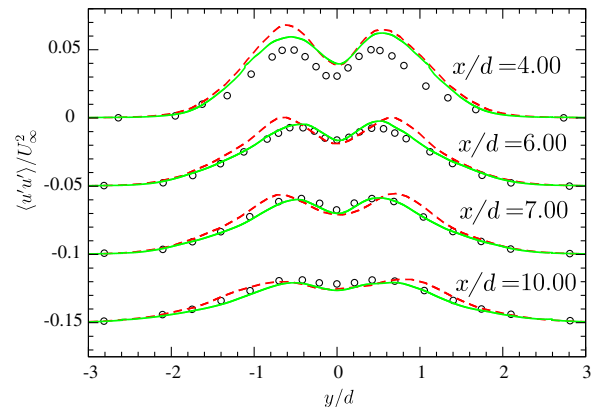


Fig. 12. Streamwise velocity fluctuations at different locations in the wake of a circular cylinder at $Re = 3900$ ($\circ$: experimental data (Ong and Wallace, 1996), --: LESOCC2 and -: INCA).

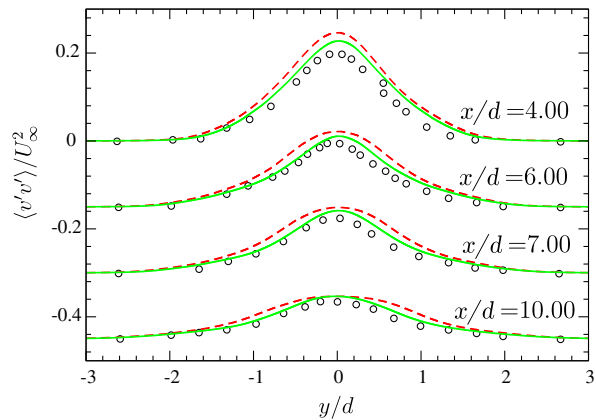


Fig. 13. Transverse velocity fluctuations at different locations in the wake of a circular cylinder at $Re = 3900$ ($\circ$: experimental data (Ong and Wallace, 1996), --: LESOCC2 and -: INCA).

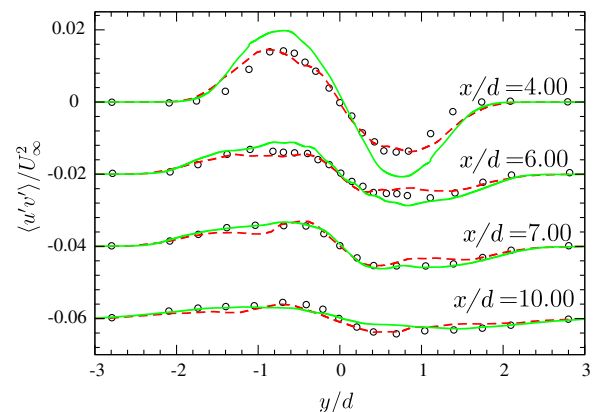


Fig. 14. Shear stress at different locations in the wake of a circular cylinder at $Re = 3900$ ($\circ$: experimental data (Ong and Wallace, 1996), --: LESOCC2 and -: INCA).

experiment is longer than the computed one. Accordingly, the calculated streamwise Reynolds stress is slightly larger than the experimental Reynolds stress (see Fig. 12). However, this finding confirms the numerical studies of Ma et al. (2000), Franke and Frank (2002) and Kravchenko and Moin (2000). Further downstream the agreement of the mean streamwise velocities and the

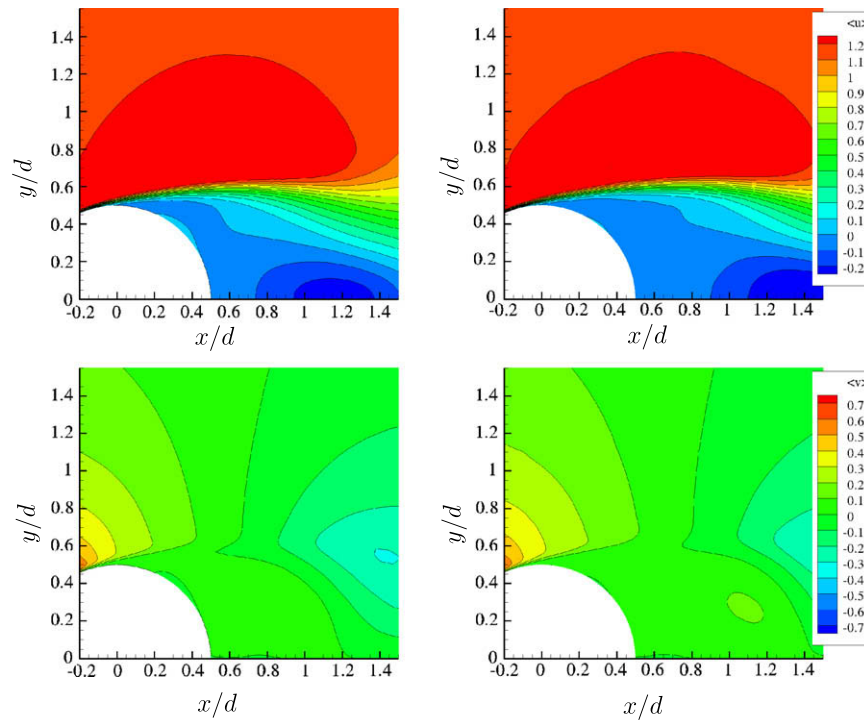


Fig. 15. Iso-contour maps of the first statistics for the flow over circular cylinder at $Re = 3900$ (LESOCC2 (left) and INCA (right): top $\langle u \rangle / U_\infty$, bottom $\langle v \rangle / U_\infty$).

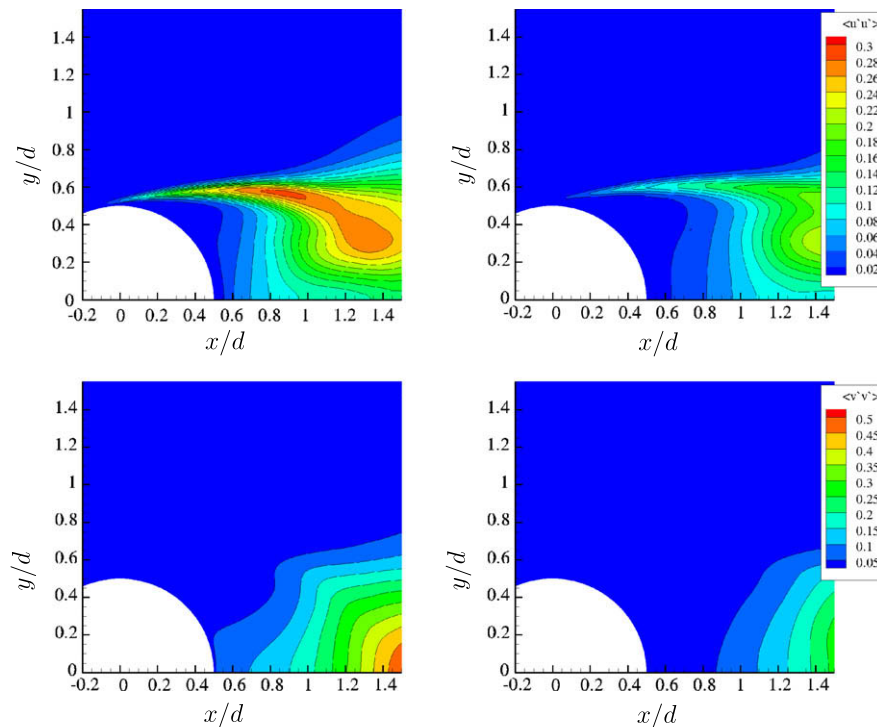


Fig. 16. Iso-contour maps of second-order statistics for the flow over circular cylinder at $Re = 3900$ (LESOCC2 (left) and INCA (right): top $\langle u'u' \rangle / U_\infty^2$, bottom $\langle v'v' \rangle / U_\infty^2$).

streamwise Reynolds stresses is good for the Cartesian-grid calculation and satisfactory for the curvilinear-grid calculation. For the cross-flow normal stress and the Reynolds shear stress, plotted in Figs. 13 and 14, there is also a slight overprediction at $x/d = 4$. Further downstream the cross-flow stresses show that the calculated wake is wider than the measured one.

3.5. One-dimensional energy spectra

In Fig. 17 the one-dimensional energy spectrum of the u -velocity component on the centerline of the wake at $x/d = 3$ is plotted. The Lomb-periodogram technique with an oversampling of 4 and a Hann window were used to perform the spectral analysis of

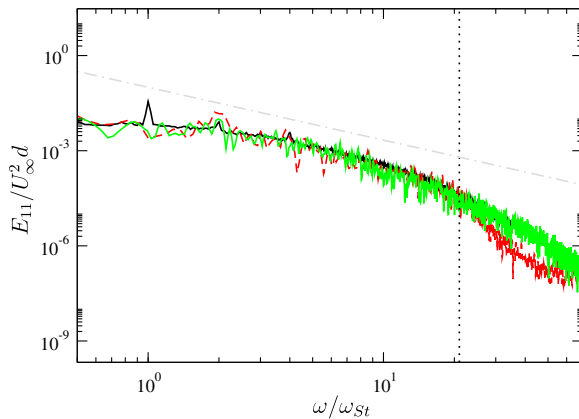


Fig. 17. One-dimensional velocity spectra at $x/d=3$ in the wake of a circular cylinder at $Re = 3900$ (— line: $-5/3$ slope, line: cut-off frequency, — with $\omega/\omega_{St} < 35$: experimental data (Parnaudeau et al., 2008), --: LESOCC2 and --: INCA).

unevenly sampled u -velocity data for the two LES simulations. About 330,000 samples of the u -velocity were collected over a time interval of $TU_\infty/d = 300$. The frequencies are normalized with the Strouhal shedding frequency ω_{St} .

Also displayed are the experimental results of Parnaudeau et al. (2008) and a $-5/3$ -slope. The spectra of the two LES and the experiment clearly show the inertial range. The good agreement for high wavenumbers for INCA with the experiment implies a correct representation down to the smallest scales. The three peaks of the u -spectra for $x/d = 3$ of the experiment are the shedding frequency and the first and second harmonic (Parnaudeau et al., 2008). The peak at $\omega/\omega_{St} = 1$ is an experimental artefact that should not be present at the centerline (Parnaudeau et al., 2008). The first harmonic can be reproduced by INCA and LESOCC2, whereas LESOCC2 also reproduces the second harmonic.

3.6. Discussion

Differences between the two calculations of the present study certainly can be attributed to the effect of CIIM in the Cartesian-grid calculation, differences in local grid resolution, and differences in the pressure treatment (staggered grid vs. collocated grid and stabilization). Fröhlich et al. (2006) also used LESOCC2 in their LES on a curvilinear grid with a dynamic Smagorinsky model and obtained for the very near wake mean velocity profiles, Reynolds stresses and respectively length of the recirculation bubble (see also Table 1) very similar results to our LESOCC2 calculation with ALDM. We emphasize that, despite these differences, results for both implicit LES are well within the range of available experimental and numerical reference data. Overall, the results of the Cartesian-grid calculation with CIIM can be considered at least as reliable as that on the curvilinear grid. The simulation with LESOCC2 consumed 3.37×10^{-6} s wall-clock time per timestep and per mesh node, while INCA consumed 1.0×10^{-6} s wall-clock time per time step and per mesh node. CIIM amounted up to 2.2% of the total wall-clock time, ALDM required approximately 2%, and the pressure Poisson solver in each case typically consumes more than 80%.

4. Conclusion

Based on Implicit Large-Eddy Simulations of the flow over a circular cylinder at $Re = 3900$ two different approaches for representing complex domain boundaries have been addressed. In the simulation code LESOCC2 body-fitted grids are employed whereas

the second simulation code, INCA, uses Cartesian grids and a recently proposed Conservative Interface Method to represent the cylinder geometry. Both simulation codes utilize the Adaptive Local Deconvolution Method (ALDM) as a subgrid-scale model. First- and second-order moments of the near wake of the circular cylinder of the two implicit LES simulations are in good agreement with each other and with experimental reference data. Mean integral quantities are well within the range of previous Large-Eddy Simulations and Direct Numerical Simulations. It can be concluded that the Conservative Immersed Interface Method and implicit LES represent the flow physics in the vicinity of the fluid–solid interface properly and are a viable alternative to body-fitted grid approaches. The straightforward generation of a locally-refined Cartesian grid and the lower computing time per grid point are the main advantages. It has been shown that the good performance of the ALDM subgrid-scale model which has been observed for canonical flows also applies to the considered flow configuration.

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